NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	09/01/2026	Issued For Review	MHP	AS	MR		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	PERUSAHAAN's Comment	Contractor's Response	Remarks

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skkmigas PERTAMINA EP	**PERTAMINA EP** **REGIONAL 1 ZONA 4**	PGG

DATA SHEET FOR ORIFICE FLOW ELEMENT (D/P TYPE)	No. Dok. : TMBP-SP001-IN-DST-D105 Rev. : A Halaman : 5 dari 8

Tag Number	: See List	Service	: See List
P&ID No.	: See List	Equip./Line No.	: See List

PROCESS CONDITION	1	Fluid		See List		
	2	State		See List		
	3	Flow	Maximum	See List		
	4		Normal	See List		
	5		Minimum	See List		
	6	Pressure	Maximum	See List	psig	
	7		Normal	See List	psig	
	8		Minimum	See List	psig	
	9	Temperature	Maximum	See List	degF	
	10		Normal	See List	degF	
	11		Minimum	See List	degF	
	12	Differential Range	Normal	See List	inH_2O	
	13		Maximum	See List	inH_2O	
	14	Density		See List	lb/ft^3	
	15	Molecular Weight		See List		
	16	Viscosity		See List	cp	
	17	Compressibility Factor		See List		
	18	Cp/Cv Specific Heat Ratio		See List		
	19	Specific Heat Ratio		See List		
	20	Specific Gravity		See List		
	21					
	22					
ORIFICE PLATE	23	Bore Type		Concentric		
	24	Standard		ISO 5167		
	25	Onfice Plate Material		316SS		
	26	Orifice Bore Diameter		See List	inch	(4)
	27	Beta Ratio		See List		(4)
	28	Min. Plate Thickness		See List	inch	(4)
	29	Vent or Drain Hole		Vent (Gas) or Drain (Liquid)		
	30					
	31					
ORIFICE FLANGE	32	Pipe Size & Bore Schedule		See List		
	33	Flange Rating & Facing		ANSI Class 150# RF		
	34	Flange Material		ASTM A105, ASME B16.5, Smooth Finished		
	35	Pressure Tap Connections		½" NPTF		
	36					
REMARKS	37	Name Plate (Tagging Number)		Note		(1)
	38	Manufacture		As per AML / PML		(2)
	39	Model No.		By VENDOR		(3)

Notes:

1. Tag No., Size and Rating, Material, Bore Diameter, Beta Ratio, and Upstream shall be stamped on orifice plate handle.
2. VENDOR shall be refer to PT. PERTAMINA HULU ROKAN Approved Manufacturers List (AML).
3. VENDOR shall submit proper model number on bid proposal.
4. VENDOR shall verify and provide the calculation.
5. Beta Ratio shall be between 0.25 and 0.7
6. VTA = VENDOR to Advise.

skkmigas PERTAMINA EP	PERTAMINA EP REGIONAL 1 ZONA 4	POG
	DATA SHEET FOR ORIFICE FLOW ELEMENT (D/P TYPE)	No. Dok. : TMBP-SP001-IN-DST-D105 Rev. : A Halaman : 6 dari 8

ORIFICE FLOW ELEMENT (D/P TYPE) LIST

No.	Tag Number	P&ID No.	Service	Equip./Line Number	Fluid	State	Pressure (Psig)		Temperature (°F)		Flow		Differential Range (inH_2O)		Density (lb/ft^3)	Molecular Weight	Viscosity (cp)	Compressibility Factor	Cp/Cv Specific Heat Ratio	Specific Heat Ratio	Specific Gravity	ORIFICE PLATE			Pipe Size & Class, Schedule
							Max	Min	Max	Min	Max	Min	Max	Min								Bore Diameter (inch)	Beta Ratio	Min. Plate Thickness (inch)	
1	FE-V-001-001	TMBP-SP001-PR-PID-D101	PROCESS LIQUID FROM PRODUCTION SEPARATOR TO WASH TANK TMB	PL-001-AA-3"	PROCESS LIQUID	LIQUID																			
2	FE-V-003-001	TMBP-SP001-PR-PID-D101	PROCESS LIQUD FROM TEST SEPARATOR TMB TO TANK TEST TMB	PL-002-AA-2"	PROCESS LIQUID	LIQUID																			
3	FE-V-002-001	TMBP-SP001-PR-PID-D102	PROCESS LIQUID FROM PRODUCTION SEPARATOR TO WASH TANK KRG	PL-003-AA-4"	PROCESS LIQUID	LIQUID																			
4	FE-V-004-001	TMBP-SP001-PR-PID-D102	PROCESS LIQUID FROM TEST SEPARATOR KRG TO TANK TEST KRG	PL-004-AA-2"	PROCESS LIQUID	LIQUID																			
5	FE-P-001-001	TMBP-SP001-PR-PID-D107	PROCESS LIQUID GROSS TRANSFER PUMP TO SP TMT	PL-085-BA-4"	PROCESS LIQUID	LIQUID																			
6	FE-FG-001-001	TMBP-SP001-PR-PID-D118	FROM TALANG JIMAR TO FUEL GAS SCRUBBER	GF-001-AA-4"	FUEL GAS	GAS																			
7	FE-F-006-001	TMBP-SP001-PR-PID-D119	FROM FUEL GAS PACKAGE TO GAS ENGINE GENERATOR	GF-003-AA-3"	FUEL GAS	GAS																			

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DATA SHEET FOR RESTRICTION ORIFICE (RO)	No. Dok. : TMBP-SP001-IN-DST-D105 Rev. : A Halaman : 7 dari 8

Tag Number	: See List	Service	: See List
P&ID No.	: See List	Equip./Line No.	: See List

PROCESS CONDITION	1	Fluid		See List			
	2	State		See List			
	3	Operation Flow	Design Flow	See List	BPD	See List	BPD
	4	Operation Temperature	Design Temperature	See List	degF	See List	degF
	5	Operation Pressure	Design Pressure	See List	psig	See List	psig
	6	Vapor Pressure		See List	psia		
	7	Discharge Pressure		See List	psig		
	8	Viscosity		See List	cp		
	9	Pipe ID		See List	inch		
	10	Specific Gravity		See List			
	11						
	12						
	13						
ORIFICE PLATE	14	Type		Note			(7)
	15	Standard		R.W Miller (ISO 5167)			
	16	Orifice Plate Material		316SS			
	17	Orifice Plate Seal Material & Type		Note			(5)
	18	Orifice Bore Diameter		See List	inch		(4)
	19	Beta Ratio		See List			(4)
	20	Min. Plate Thickness		See List	inch		(4)
	21	Calculate Noise		Note			(4)
	22	Vent or Drain Hole		N/A			
	23						
ORIFICE FLANGE	24	Pipe Size & Bore Schedule		See List			
	25	Flange Rating & Facing		See List			
	26	Flange Material		Refer to piping material specification			
	27	Pressure Tap Connections		N/A			
	28						
REMARKS	29	Name Plate (Tagging Number)		Note			(1)
	30	Manufacture		As per AML / PML			(2)
	31	Model No.		By VENDOR			(3)
	32						
	33						
	34						
	35						
	36						
	37						
	38						
	39						

Notes:

1. Tag No., Size and Rating, Material, Bore Diameter, Beta Ratio, and Upstream shall be stamped on orifice plate handle.
2. VENDOR shall be refer to PT. PERTAMINA HULU ROKAN Approved Manufacturers List (AML).
3. VENDOR shall submit proper model number on bid proposal.
4. VENDOR shall verify and provide the calculation.
5. VTA = VENDOR to Advise.
6. Process data refer to Hysys data
7. All tag no. is single plate except RO-P005-002, RO-P005-002 will use multistage restriction orifice type

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DATA SHEET FOR RESTRICTION ORIFICE (RO)

No. Dok. : TMBP-SP001-IN-DST-D105
Rev. : A
Halaman : 8 dari 8

RESTRICTION ORIFICE LIST

No.	Tag Number	P&ID No.	Service	Equip./Line Number	Fluid	State	Pressure (Psig)		Temperature (°F)		Flow (BDP)		Vapor Pressure (Psig)	Discharge Pressure (Psig)	Viscosity (cp)	Pipe ID	Cp/Cv Specific Heat Ratio	ORIFICE PLATE			Pipe Size & Class, Schedule
							Max	Min	Max	Min	Max	Min						Bore Diameter (inch)	Beta Ratio	Min. Plate Thickness (inch)	
1	RO-P-002-001A	TMBP-SP001-PR-PID-D103	TEST TANK TMB AND TEST PUMP TMB	PL-011-AA-2"	GROSS	LIQUID	39.0	59.0	100.68	150.68	2070	2300	-14.70	0.15	1.050	2"	1.140				2"/150#/40S
2	RO-P-003-001A	TMBP-SP001-PR-PID-D104	TEST TANK KRG AND TEST PUMP KRG	PL-020-AA-2"	GROSS	LIQUID	39.0	59.0	80.68	130.88	2200	2500	-14.70	0.15	59.12	2"	1.140				2"/150#/40S
3	RO-P-001-001A	TMBP-SP001-PR-PID-D107	GROSS TRANSFER PUMP	PL-085-BA-4"	GROSS	LIQUID	174.65	194.65	81.45	131.00	11925	13195	-14.70	0.15	1.11	4"	1.150				4"/150#/40S
4	RO-V-005-001A	TMBP-SP001-PR-PID-D114	PRODUCTION SCRUBBER	PL-005-AA-2"	PROCESS LIQUID	LIQUID	28.97	49.0	79.86	129.86	3	5	-14.70	0.5	0.01	2"	1.240				2"/150#/40S
MINIMUM FLOW																					
5	RO-P-002-001B	TMBP-SP001-PR-PID-D103	TEST TANK TMB AND TEST PUMP TMB	PL-011-AA-2"	GROSS	LIQUID	39.0	59.0	100.68	150.68	621	683	-14.70	0.15	1.050	2"	1.140				2"/150#/40S
6	RO-P-003-001B	TMBP-SP001-PR-PID-D104	TEST TANK KRG AND TEST PUMP KRG	PL-020-AA-2"	GROSS	LIQUID	39.0	59.0	80.88	130.88	660	726	-14.70	0.15	59.12	2"	1.140				2"/150#/40S
7	RO-P-001-001B	TMBP-SP001-PR-PID-D107	GROSS TRANSFER PUMP	PL-085-BA-4"	GROSS	LIQUID	572.5	592.50	84.57	134.87	3599	3958	-14.70	0.15	1.11	4"	1.150				4"/150#/40S
8	RO-V-005-001B	TMBP-SP001-PR-PID-D114	PRODUCTION SCRUBBER	PL-005-AA-2"	PROCESS LIQUID	LIQUID	29.0	49.0	79.86	129.86	3	5	-14.70	0.5	0.01	2"	1.240				2"/150#/40S